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Division of Dockets Management
U.S. Food and Drug Administration
5630 Fishers Lane
Room 1061 (HFA-305)
Rockville, MD 20852

Re: Draft Guidances Relating to the Regulation of Human Cells, Tissues,
and Cellular and Tissue-Based Products
(Docket No. FDA-2015-D-3719)

Dear Sir or Madame:

Thank you for the opportunity to comment on Docket No. FDA-2015-D-3719, Draft Guidances Relating to the Regulation of Human Cells, Tissues, and Cellular and Tissue-Based Products.

As a former FDA Associate Commissioner and practicing food, drug and device regulatory lawyer, I have worked with hundreds of clients on constructive ideas that will help advance medical innovation. Alston & Bird and I also represent the Bipartisan Policy Center on matters associated with advancing medical innovation, including informing expert panel recommendations that provide constructive input to help modernize the Federal Food Drug and Cosmetic Act (FDCA) and create a practical statutory pathway tailored to the unique attributes of cell and tissue-based therapies.

My colleagues and I have represented dozens of clients seeking Agency review and approval of human cells, tissues, or cellular or tissue-based products (HCT/PS). Some of these clinicians, or product manufacturers, sought qualification under 21 CFR Part 1271 for products for which FDA's primary public health concern was to prevent the introduction, transmission, and spread of communicable disease.¹ Others have

¹ After many years of internal analysis and debate within a Tissue Reference Group organized by FDA to recommend a regulatory solution to the use of human cells and tissues to treat disease which, like organs, were not specifically regulated by FDA, the Agency used §361 of the Public Health Service Act (PHA) as its statutory authority to promulgate regulations to bring these previously unregulated components within FDA

encountered inconsistent or unnecessarily restrictive agency interpretations of the key definitions in the Part 1271 regulations such as “autologous,” “homologous,” “more than minimally manipulated” and what constitutes the “same surgical procedure.” Therefore, from the perspective of clarifying discretion or ambiguity in the application of these terms, and promoting consistency, the draft guidance from the Agency is welcome and appreciated. However, in issuing this draft guidance, FDA adopted language and examples that are even more restrictive and conservative than its actual application of these rules in review of existing products.

In every case in which FDA determined, through a designation request, that a product did not qualify for regulation solely under §361, and where the only alternative, a full-Biologics Licensing Application (BLA) or device Premarket Approval Application (PMA) was required, the sponsors (and our clients) dropped the projects for various reasons. These products were never developed commercially. Where FDA evaluated the technology and interpreted these §1271 definitions more flexibly, the products were funded, developed, manufactured and led to improvements in public health.

Surprisingly, none of these more flexible interpretations were included as examples in the draft guidance. These real-life examples, which FDA is in the best position to summarize, could encourage innovation and use of accepted regenerative technologies. Only the most obvious and restrictive examples were included in the guidance documents (discussed in the Outline of Comments attached). These draft guidances, if finalized as written, essentially shut down emerging uses of various types of stem cells; intentionally channeling all such uses into conventional drug development requirements, including INDs, IDEs, three phases of human clinical research programs developed and funded to support submission of full BLAs and PMAs 10-15 years later at a cost of between \$200 million - \$1 billion in commercial development costs. The problem is, of course, that in over 5,000 reported on-going human clinical trials of HCT/Ps underway pursuant to approved INDs or IDEs in the last 10-years, FDA has, to our knowledge, received no data to review through filed BLAs or PMAs.

jurisdiction. The basis of §361 (42 U.S.C. §264(a)) was narrowly to prevent the introduction, transmission and spread of communicable diseases. In a subsequent legal test, *U.S. v. Regenerative Sciences, LLC*, 878 F. Supp.2d 248 (D.C.D.C. 2012), 741 F.3d 1314 (D.C. Cir. 2014), the Federal courts upheld FDA’s authority to regulate HCT/Ps directly as drugs and biologics under §351 PHA (e.g., autologous MSC cells extracted and isolated from bone marrow, cultured to divide and proliferate, combined with the antibiotic doxycycline to eliminate any contamination, and reinjected to treat orthopedic diseases). The courts held that these cell-based products were drugs/biologics that FDA could regulate under §351 PHA. Of course, that decision did not state that FDA had to regulate these HCT/Ps under traditional drug pathways, or that the Regenexx™ Procedure did not work in repairing orthopedic injuries.

An example of the flexible application of the Part 1271 rules is the use of amniotic membranes to treat corneal erosion which FDA has reportedly allowed. Yet this more flexible application of the homologous definition, and other more far reaching uses, was not listed as an example to guide regulatory professionals and make development resources available. The only examples included in the draft guidance for homologous use involve the most obvious situations including skin grafts using donor skin cells identical to recipient skin cells, or use of a donor heart valve to replace the same heart valve in the recipient. More interesting, dynamic and “edge of the envelope” examples are required within the actual experience of FDA in order to highlight these lines and make these guidances more useful and relevant.

Likewise, FDA seems to confuse the plain meaning of the “minimal manipulation” definition in §1271.3(f)(2) for cells (e.g., “processing that does not alter the relevant biological characteristics of cells”) by requiring these cells also meet the separate requirements of the structural tissue from which cells are retained as defined in §1271.3(f)(1)(e.g., that processing “does not alter the original relevant characteristics of the tissue relating to the tissue’s utility for reconstruction, repair or replacement.”) Isolating or removing stromal/stem cells from adipose tissue, for example, is determined to be more than minimally manipulated, therefore requiring a full-BLA for any use because, according to FDA, if “the processing breaks down and eliminates the structural components that provide cushioning and support, thereby altering the original relevant characteristics of the HCT/P relating to its utility for reconstruction, repair or replacement.”² Arguably, if the product under development uses isolated stem cells from adipose tissue, the processing of those cells should be determined based on the relevant biological characteristics of the cells themselves (e.g., ability to proliferate, differentiate, etc.), and not the characteristics of the adipose fat tissue from which those cells are harvested. The fat tissue is not used in the manufacture or processing of the isolated and purified MSC stem cells.

These two specific examples highlight important short comings in the draft guidance. Additional comments related to each guidance are summarized in the attached Outline of Comments attached to this correspondence. Our most significant comments include the following:

A. Minimal Manipulation Draft Guidance

- FDA should view extraction or separation of cells independently from the underlying function of the structural tissue, where the cells are the intended product and not the tissue from which they are harvested.

² Example 10-1, pg. 8, *Guidance- Minimal Manipulation of Human Cells, Tissues, and Cellular and Tissue-Based Products*.

- The §1271 definition of “minimal manipulation” (MM) provides FDA with this important tool to evaluate cells independently as minimally manipulated if processing “does not alter the relevant biological characteristics of cells” (1271.3(f)(2)).
- It should be irrelevant scientifically and legally if, for example, use of these cells is not related to the relevant characteristics of fat tissue to pad and cushion against shocks (the tissue and not the cell criteria), if the cells have the inherent biologic characteristics of being able to differentiate and proliferate when isolated—the value of stem cells in restorative medicine.
- Limiting the allowable use of stromal/stem cells to the characteristics of the tissue in which they are stored is contrary to the language of the regulation and its format which includes a separate definition of minimal manipulation for cells and structural tissue.
- The cell definition should be used if cells are the intended product (e.g., does processing alter the relevant biological characteristics of the cells). The structural tissue definition should be used if the tissue is the intended product (e.g., does processing alter the tissue’s utility, for reconstruction, repair or replacement).
- The Guidance should focus on whether the extraction, processing, separation, or other manipulation of the tissue/cells increases the risk of infection or introduction of substances with unproven safety profiles into the patient.
- FDA seems to intentionally eliminate the utility of stromal/stem cells under §1271 regardless of the techniques used to isolate and process those cells by requiring that extraction or separation of cells from structural tissue is considered more than minimal manipulation (MMM) if removal reduces the bulk of the adipose (fat) tissue from which they are extracted (Guidance, pg. 8, Example 10-1).
- This interpretation is contrary to the language and intent of the regulatory definition of minimal manipulation.
- In Section C of the draft guidance, FDA differentiates between obtaining a higher concentration of hematopoietic stem/progenitor blood cells using a mobilized peripheral blood apheresis product which it deems minimally manipulated, and incubating selected placental/umbilical cord blood cells in a laboratory vessel containing culture media and/or growth factors to achieve a large number of cells capable of long-term repopulation of bone marrow as more than minimally manipulated (because the processing theoretically may affect production of intracellular or cell-surface proteins and other markers) which likely have little or no significance in the clinical outcome.

- The science in the field, or the regulatory definition of minimal manipulation, does not require FDA to take such a narrow or restrictive “either-or” approach. FDA could more generally determine simply that the relevant biological characteristics of cells include the ability to divide, expand and proliferate. Since cells contain the ability to reproduce and proliferate, allowing them to do so naturally in a supportive medium arguably does not alter their relevant biologic characteristics to divide or proliferate.
- Simply taking a small step back in its regulatory interpretations, when justified by established existing science, could make use of these blood stem cells much more effective, for example, in treating blood cancers, and improve public health.

B. Homologous Use Draft Guidance

- The regulatory definition of “homologous use” that allows application of §1271 is broad (e.g., use of cells that perform the same basic function or functions in the recipient as in the donor). Again, the cited examples referenced by FDA are unnecessarily narrow.
- Infusion of hematopoietic stem cells to treat blood disorders is given as an example of homologous use, but infusion intravenously to differentiate into neuronal cells to treat cerebral palsy is given as an example of non-homologous use even though those cells have this demonstrated capability. (Actually the use of cord blood cells for CP is not based on differentiation into healthy nerve cells but documented paracrine signaling to reduce inflammation and promote neural connectivity.) If these cells possess the biological characteristics to proliferate, differentiate and reduce inflammation, FDA has the discretion to accept evidence that these blood borne stem cells perform the same function in the donor as in the recipient.
- Less obvious examples should be included in the guidance where FDA has permitted tissue or stem cell uses as homologous other than the most obvious ones included; e.g., highlight the uses of amniotic cells or tissues in the eye to treat corneal defects, ocular inflammation (*see* www.biotissue.com), or wound healing, rather than limit examples to skin grafts or heart valve replacements.

C. Homologous Use of Adipose Tissue Guidance

- Issuance of this guidance specific to use of adult stem cells derived from fat tissue seems designed to “slam the door” on any continued use of cell therapy under §1271 without submission and approval of an FDA premarket approval application.
- The guidance notes that processing to isolate non adipocyte or non-structural components from adipose tissue (with or without subsequent cell culture or

- expansion) is MMM because it alters the original relevant characteristics relating to the tissue's utility for reconstruction, repair or replacement (Guidance at pg. 3).
- As noted in the discussion of the Minimal Manipulation Guidance above, FDA again intentionally combines the MM definition of cell processing with processing of the underlying structural tissue which is not part of the product being developed. This runs completely counter to the construction of the regulatory language of §1271.3(f).
 - It is legally untenable and a misreading of the regulation (§1271.3(f)(2) v. (f)(1)) to deem MSCs isolated from adipose tissue as MMM based on the characteristics of the tissue from which these cells are removed. The standard for MM of cells is whether the processing alters the relevant biological characteristics of the cells ((f)(2)), not whether the processing alters the tissue's utility for reconstruction, repair or replacement ((f)(1)).
 - Use of adipose tissue or adipose-derived MSC for breast augmentation should be considered homologous, not used as an example of non-homologous use because of the narrow determination that breast tissue is not fat tissue, but tissue to produce milk (lactation) after childbirth.
 - The draft guidance states that if stem cells are isolated from lipoaspirate, the adipose tissue would no longer be considered HCT/P. Therefore, in FDA's view, even if these cells are injected into the same patient, the "same surgical procedure exception" under §1271.15(b) would not apply (Guidance at pg. 8).
 - Again, FDA confuses the manipulation and use of the cells under §1271.3(f)(2) from use of the underlying tissue under §1271.3(f)(1).
 - There is no legal authority, of which we are aware, to determine that stromal MSC or other cells derived from adipose tissue are not HCT/Ps. HCT/Ps are defined broadly as "articles containing or consisting of human cells or tissues that are intended for implantation, transplantation, infusion, or transfer into a human recipient" (1271.3(d)).

D. Same Surgical Procedure Guidance

- The Guidance is intended to clarify the application of §1271.15(b) that exempts procedures from §1271 if an establishment removes HCT/Ps from an individual and implants "such HCT/Ps" back into the same individual during the same surgical procedure.
- The Guidance, not the regulation, interprets "such HCT/Ps" to mean that they are in their original form (Guidance, footnote 4, pg. 3). This interpretation deviates from the MM standard with FDA stating that the HCT/P can only be "rinsed,

cleaned, sized or shaped” to qualify for the same surgical procedure exception, and cells cannot be removed from host tissue.

- This discussion represents cynical, unsupported legal gymnastics creating inconsistencies between the MM standard that permits extraction and isolation of stem cells from underlying tissue by further denying use of such cells without drug or device approval even if autologous, homologous, MM stem cells are implanted in the same surgical procedure.
- The limited examples involve only tissue transfers not cells (e.g., skin grafting and coronary bypass surgery)(Guidance at pg. 4). Examples of the same surgery exception should be applied to a narrow use of stem cells (e.g., hematopoietic stem cell transplants to treat blood borne disease, etc.).

E. Stem Cell Clinics

The important question has been raised concerning the impact of unapproved product uses of stem cells under §1271 by unapproved or unregistered U.S. clinics. Recent “back of the envelope” estimates based on Google searches have been included in on-line blogs. This analysis has estimated that as many as 351 businesses are engaged in direct-to-consumer marketing of stem cell interventions offered by 570 clinics throughout the U.S.³

FDA certainly maintains the existing regulatory power to inspect and issue Warning Letters, or take other compliance action, against clinics or other health providers engaged in promoting or manufacturing unapproved drugs or devices.⁴ The Agency has issued a series of high visibility Warning Letters to stem cell clinics including those that recover and process adipose tissue from donors for autologous use.⁵ Many of those letters relate to clinics offering stem cell therapy to treat a variety of diseases and conditions including: autism, Parkinson’s disease, pulmonary fibrosis, chronic obstructive pulmonary disease (COPD), multiple sclerosis (MS), cerebral palsy and amyotrophic lateral sclerosis

³ L. Turner and P. Knoepfler, *Cell Stem Cell*, Vol 19, Issue 2, pg. 154-157, Aug. 4, 2016.

⁴ See §§501(a) FDCA concerning sale of adulterated drugs and devices, 502(a) misbranded drugs and devices and the 301(a) and (b) making it a “prohibited act” to introduce into interstate commerce any adulterated (unapproved) or misbranded drug or device.

⁵ See, for example, Warning Letter to Irvine Stem Cell Treatment Center, December 30, 2015 (<http://www.fda.gov/ICECI/EnforcementActions/WarningLetters/2015/ucm479837.htm>)

(ALS).⁶ The vast majority of these clinics, however, according to recent surveys, relate to use of stem cells to treat more common orthopedic injuries and cosmetic surgery.

Certainly, FDA does not possess the capacity, or wish to use its limited compliance and enforcement resources, to seek closure of hundreds of medical practices, while others open in their place. The proliferation of existing medical clinics offering stem cell therapies should represent a huge “red flag” to FDA that something in the existing regulation of HCT/Ps is not working and requires modification. Consumer demand is increasing quickly while few, if any, cell-based products are advancing through the traditional drug and device application process.

We recommend that compromises be considered in the context of finalizing these guidance documents that more clearly prioritize and identify cell uses that represent a reasonable interpretation of Part 1271. FDA should identify practices that it will prohibit even if the alternative is not realistic; that a sponsor will pursue a full BLA or PMA. Brighter lines need to be established to permit reasonable medical procedures in the clinic in areas where safety has been established, and risks are small, in order to prioritize FDA enforcement resources, solidify legal standing and convince medical professionals to comply with the existing regulatory regime.

Thank you for the opportunity to comment and for your thoughtful consideration.

Sincerely,

A handwritten signature in blue ink, appearing to read "Marc J. Scheineson".

Marc J. Scheineson

enclosure (Outline of Comments FDA Draft Guidance (HCT/Ps)
LEGAL02/36667590v1

⁶ *Id.*